INCH-POUND

MIL-M-70696A(AR)
15 September 1993
SUPERSEDING
MIL-M-70696(AR)
10 July 1988

MILITARY SPECIFICATION

MINE ASSEMBLY, ANTI-PERSONNEL, M86
LOADING AND ASSEMBLING OF

This specification is approved for use by the U.S. Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading and assembling of the Mine, Antipersonnel M86 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation. (see 6.2)

SPECIFICATIONS

MILITARY

MIL-E-14970	- Explosive Composition A5.
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For.
MIL-D-48183	- Delay Detonator
MIL-M-48245	- Molding Compound, Housing and Plug.
MIL-P-48333	- Propellant, Liquid.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1345

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STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-286 - Propellant, Solid Sampling, Examination and Testing.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (6.11)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9366700 - Mine Assembly, M86.
9366710 - Packing and Marking for Box, Wirebound for M86 APERS Mine with Bandoleer.
9298699 - Delay Detonator Functional Test Configuration.
12910121 - M86 APERS Mine with Bandoleer.

(Copies of other Government documents, drawings, publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D695-85 - Plastic Rigid, Compressive Properties of.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Mine. The mine shall comply with all requirements specified on Drawing (dwg.) 9366700 and with all requirements specified in applicable specification.

3.3 Documentation of controls for use of adhesives and sealants. The contractor shall develop a documented operational procedure for utilization of adhesives and sealants. This procedure shall provide for control of the materials and processes to be used; including controls for the following: shelf life, handling, mixing, curing and pot life. Necessary cleaning agents, safety precautions, ventilization, and any other information considered necessary for the proper use of the adhesive(s) or sealant(s) shall also be included. The procedure shall be updated whenever a change is made to the process such that product quality may be affected.

3.4 Static functioning test of kill mechanism and overlay assembly. The kill mechanism and overlay assembly shall function satisfactorily as evidenced by compliance with all of the following:

a. The test housing containing the kill mechanism and overlay assembly (K.M.&O.) shall be broken and entirely separated from the kill mechanism assembly.

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b. The kill mechanism assembly shall be projected upward from the test housing and shall explode its main charge high order, i.e., no kill mechanism fragment shall be larger than 0.25 inches in any direction.

c. The average height at which the main charge explodes shall be not less than 2.5 feet above the surface upon which the housing was resting.

Only those lots of kill mechanism and overlay assemblies which have met the acceptance criteria in Table II for the static functioning test shall be assembled into the molded assembly.

3.5 Qualification test of delay detonator and propellant nut assembly serial lots. The delay detonator serial lot (MIL-D-48183) and propellant nut assembly serial lot to be used in the kill mechanism and overlay assembly lot shall perform satisfactorily when tested together as evidenced by compliance with the qualification test criteria specified in 4.5.5.

3.6 Documentation of plan for safety in handling and processing of munitions. Prior to any loading, the contractor shall document a detailed written plan of the procedures to be employed during handling of the munitions. This documentation shall describe as a minimum, procedures to be employed to assure that:

a. The shorting bar is not deformed or broken.

b. The battery ball is not depressed.

c. The liquid propellant does not leak externally from overlays.

d. The Safe & Arming (S&A) safety pin is not deformed, broken or missing.

e. The temperature of the molded assembly or munition does not exceed 155°F at any time during storage or processing.

f. HE and inert munitions remain properly identified and segregated at all times.

g. The S&A safety pin (temporary) is present and properly positioned in each molded assembly prior to loading.

Submission of this plan shall be in accordance with Data Item Description. Any changes to the approved plan shall be submitted for approval, prior to implementation.

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3.7 Bulk density, moisture content and consolidation pressure of explosive composition A5. The contractor shall provide a means of automatically and continually monitoring and recording the consolidation pressures on each machine used in loading the upper and lower shell assemblies and shall adjust charge weights based on the recorded pressures, such that the consolidation density remains constant with volume changes in the lots of shell metal parts. The bulk density of the explosive composition A5 loaded into the upper and lower shell assemblies shall be not less than 0.83 nor more than 1.00 grams/cc as determined by the "Bechtel Method". The moisture content of the explosive composition A5 shall not exceed 0.10 percent at the time of loading.

3.8 Documentation of plan for molding and cure process controls. The contractor shall prepare a written plan to be implemented for control of the process for molding the kill mechanism and overlay assembly into the housing, timing, and fuzing. The plan shall briefly describe the entire process and shall fully document the controls to be implemented for maintaining the process. The controls shall include, as a minimum, all the checkpoints referenced in 6.4. Submission of this plan shall be in accordance with Data Item Description.

Approval of the plan shall be limited to the controls to be employed and not the overall potting process, nor shall such approval relieve the contractor, in any way, of the responsibility for producing acceptable product.

3.9 Molding compound for casting into molded assemblies. The molding compound used for casting the housing shall comply with MIL-M-48245. The resin/hardener weight ratio shall be 2.0/1.0 $\pm$ 5% (1.8/1 to 2.2/1) and the compressive strength shall be 25,000 p.s.i. minimum. The color of the arc face on the completed munition shall be within the range of colors bounded by color numbers 34079 and 34102 inclusive of FED-STD-595.

3.10 Temperature control during molding and cure of munition. The Molded Assembly, or Munition shall not be subjected to temperatures exceeding 155°F at any time.

3.11 Leakage of liquid propellant. There shall be no evidence of liquid propellant leakage in the molded assembly from the Kill Mechanish (KM) and overlay assembly before or after loading.

3.12 Functioning.

3.12.1 Safe dud. The mine, if not functioned by any external or internal mode, shall not function after the maximum self-destruct time has elapsed.

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3.12.2 Hazardous dud. All safe duds when tested as specified in 4.5.5.1, shall not be capable of functioning after the maximum self-destruct time has elapsed.

3.13 Data requirements. The contractor shall generate data in accordance with the requirements of the data item descriptions cited in 3.3, 3.6, 3.8, 4.4.2, 4.4.2.21, 4.5.14.6, 6.8, 6.10.

3.14 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.15 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, discoloration and other defects which would make the item unsuitable for its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection.

b. Quality Conformance Inspection.

c. Lot Acceptance Testing.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Propellant Nut Assembly	9298433	50
Overlay Assembly	9298412	25
Kill Mechanism and Overlay Assembly, HE	9298405	25
Mine Assembly, M86	9366700	36

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See Table II and MIL-A-48078.

If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon failure of any assembly, component or test specified in the sample to comply with any of the stated requirements.

NOTE: The Government reserves the right to terminate its inspection when any failure of an assembly, component or test specimen in the sample causes rejection per the accept-reject criteria for that examination or test.

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Table I - First article inspection

CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Mine Assembly, M86	1 OF 1	See Below
			NEXT HIGHER ASSEMBLY: ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Propellant Nut Assembly (Dwg. 9298433)			
	Examination for defects	50	3.2	4.4.2.6
	Qualification Test	50	3.2	4.5.5
	Overlay Assembly (Dwg. 9298412-1)			
	Examination for defects	25	3.2	4.4.2.19
	Leakage Test	25	3.2	4.5.9
	Kill Mechanism and Overlay Assembly, HE (Dwg. 9298405)			
	Examination for defects	25	3.2	4.4.2.21
	Leakage Test	25	3.2	4.5.10
	Mine Assembly, M86 (Dwg. 9366700)			
	Examination for defects	36	3.2	4.4.2.26
	Bag Assembly (Dwg. 12555278)			
	Examination for defects	24	3.2	

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of mines shall contain:

a. Molded Assembly, PDM of one lot interfix number from one manufacturer.

b. Shell tube assemblies (8865570) of one lot interfix number from one manufacturer.

c. Delay detonators (9298429) of one lot interfix number from one manufacturer.

d. Overlay assemblies from one lot interfix number from one manufacturer.

e. Detonating cord assemblies (9298406) of one lot interfix number from one manufacturer.

f. Explosive composition A5 from one batch or blending of batches which results in homogeneity of all composition used in loading the upper and lower shell assemblies.

g. Propellant (9298437) and WC-140 from one batch or blending of batches which results in homogeneity of all propellant used in loading the propellant nut assembly.

h. Propellant nut assemblies from one lot.

i. Liquid propellant from one batch or blending of batches which results in homogeneity of all composition used in loading the overlay assemblies.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table I below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE II Attributes sampling inspection.

Lot Size	Inspection Level I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13

Numbers under inspection levels indicate sample size; asterisks (*) indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

Lot Size	Special Inspection Levels S-1	S-II	S-III	S-IV	S-V	S-VI
2 to 15	*	*	*	13	5	3
16 to 50	*	*	32	13	5	3
51 to 150	*	125	32	13	5	3
151 to 500	*	125	32	13	5	3
501 to 3200	1250	125	32	13	13	5
3201 to 35000	1250	125	32	13	13	5
35001 to 500000	1250	125	32	13	20	8
500001 and over	1250	125	32	32	20	8

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.5). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

4.4.3 Examination. See MIL-A-48078.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Nut, Propellant	1 OF 1	9298434
			NEXT HIGHER ASSEMBLY 9298433

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Hole missing or blocked	Level II	3.2	Visual
102	Diameter of hole	Level II	3.2	Gage
103	Location of hole from center of nut	Level II	3.2	Gage
104	Width of groove	Level II	3.2	Gage
105	Depth of groove	Level II	3.2	Gage
106	Flange thickness	Level II	3.2	Gage
107	Distance from shoulder face to bottom of small diameter cavity	Level II	3.2	Gage
108	Distance from shoulder face to bottom of large diameter cavity	Level II	3.2	Gage
109	Diameter of small cavity	Level II	3.2	Gage
110	Diameter of large cavity	Level II	3.2	Gage
111	Major diameter of thread	Level II	3.2	Gage
112	Pitch diameter of thread	Level II	3.2	Gage
Minor				
201	Diameter of flange	Level IV	3.2	Gage
202	Poor workmanship	Level IV	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Plate, Orifice	1 OF 1	9298435
			NEXT HIGHER ASSEMBLY: 9298433

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Hole missing or blocked	Level II	3.2	Visual
102	Diameter of hole	Level II	3.2	Gage
Minor				
201	Diameter	Level IV	3.2	Gage
202	Poor workmanship	Level IV	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH 4.4.2.3	TITLE Propellant (Prior to Loading into Propellant Nut)	SHEET 1 OF 1	DRAWING NUMBER 9298437
			NEXT HIGHER ASSEMBLY 9298433

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Propellant moisture incorrect		3.2	Certification Notes 1, 2
102	Moisture content of propellant		3.2	4.5.3
Minor				
201	Poor workmanship	Level III	3.15	Visual

NOTES:

1. Certification shall include the actual percentage by weight of each constituent immediately prior to blending.
2. The propellant mix shall be lotted and a Propellant Description Sheet (RRCOM-234-R) shall be prepared for each lot. The sheet shall be available for review by the Government.

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH 4.4.2.4	TITLE Propellant Nut (Prior to Loading of Propellant)	SHEET 1 OF 1	DRAWING NUMBER 9298434
			NEXT HIGHER ASSEMBLY 9298433

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Thread damaged	Level II	3.2	Visual
102	Foreign matter in any cavity or hole	Level II	3.2	Visual
103	Groove missing	Level II	3.2	Visual
104	Seal at base of propellant cavity missing	Level II	3.2	Visual
Minor				
201	Poor workmanship	Level III	3.15	Visual

NOTES:

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Propellant Nut Assembly (Prior to Assembly of Orifice Plate)	1 OF 1	9298433
			NEXT HIGHER ASSEMBLY 9298433

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Weight of propellant		3.2	NOTE A
Minor 201	Poor workmanship	Level III	3.15	Visual
NOTE A:	The Propellant Nut Assembly machine will be qualified at the beginning of each production interval by processing a minimum of two (2) acceptable weight samples prior to use in production. The machine will be requalified each time it is removed from the production process for maintenance or adjustment. A minimum of two (2) samples shall be processed each hour of production or fraction thereof. Each hour's production or fraction thereof shall be accepted provided all samples processed during that period meet weight requirements. If a sample fails to meet weight requirements, all loaded propellant nut assemblies processed since the last acceptable check shall be rejected. The assembly machine shall be stopped for corrective action and requalified prior to release for production.			

NOTES:

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Propellant Nut Assembly	1 OF 1	9298433
			NEXT HIGHER ASSEMBLY: 9298433

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Propellant present on threads	SI	3.2	Visual
b	Propellant present on other surface outside of the propellant cavity	SI	3.2	Visual
Major				
101	Seal under orifice plate missing	Level II	3.2	Visual
102	Stake of orifice plate missing (4 places) or 360° crimp	Level II	3.2	Visual
103	Foreign matter or burr in hole of orifice plate	Level II	3.2	Visual
104	Force test of stake	Level II	3.2	4.5.4
105	Qualification test of propellant nut assembly and delay detonator	Note 1	3.5	4.5.5/9298699
Minor				
201	Poor workmanship	Level III	3.15	Visual

NOTES:
1. The samples shall be selected from the machine at the beginning of production, and equal intervals during production of the lot and from the last portion of production of the lot.

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Explosive Composition A5, Prior to loading	1 OF 1	9298357
			NEXT HIGHER ASSEMBLY: 9298357

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Bulk density		3.7	4.5.2.1
102	Moisture content		3.7	4.5.1
Minor				
201	Poor workmanship	Level III	3.15	Visual

NOTES:

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Upper Shell Assembly	1 OF 1	9298357
			NEXT HIGHER ASSEMBLY: 9298417

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Presence of sleeve, outer, positioned over tube assembly	Level II	3.2	Visual
Major				
101	Sleeve height	Level II	3.2	Gage
102	Depth to charge	Level II	3.2	Gage
103	Outside diameter, max.	Level II	3.2	Gage
104	Diameter of detonator tube sleeve, min.	Level II	3.2	Gage
105	Inside diameter of detonator tube	Level II	3.2	Gage
106	Charge chipped, cracked or pitted	Level II	3.2	Visual
107	Density of charge	Level II	3.2	4.5.2
108	Spring grip fastener missing	Level II	3.2	Visual
109	Consolidation pressure	Level II	3.7	Certification
110	Presence of exuded adhesive (360°) at A5 interface (presence of RTV (360°) on top of sleeve)	3.2	Visual	
111	Outersleeve secure	Level II	3.2	Finger tight
Minor				
201	Poor workmanship	Level III	3.15	Visual

NOTES:

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MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Lower Shell Assembly	1 OF 1	9298358
			NEXT HIGHER ASSEMBLY: 9298417

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Depth to charge	Level II	3.2	Gage
102	Outside diameter, max.	Level II	3.2	Gage
103	Diameter of hole in shell	Level II	3.2	Gage
104	Diameter of hole in charge	Level II	3.2	Gage
105	Charge chipped, cracked or pitted	Level II	3.2	Visual
106	Density of charge	Level II	3.7	Certification
108	Comp. as shoulder 360° not present	Level II	3.2	Visual
Minor				
201	Poor workmanship	Level IV	3.15	Visual

NOTES:

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MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Delay Detonator	1 OF 1	9298429
			NEXT HIGHER ASSEMBLY 9298417

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Delay composition not below flush	Level II	3.2	Visual
102	Delay composition cracked or chipped	Level II	3.2	Visual
103	Qualification test	Level II	3.5	4.5.5/9298699
Minor				
201	Poor workmanship	Level III	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Kill Mechanism Assembly	1 OF 1	9298417
			NEXT HIGHER ASSEMBLY: 9298405

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter, max.	Level II	3.2	Gage
102	Total height, max.	Level II	3.2	Gage
103	Foreign matter on exterior surface (explosive, excessive sealant, etc)	Level II	3.2	Visual
104	Sealant missing on any portion of joint between upper and lower shell assemblies	Level II	3.2	Visual see 4.5.6
105	Torque of propellant nut assembly	Level II	3.2	4.5.7
Minor				
201	Poor workmanship	Level IV	3.15	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Seal, Upper Shell (Prior to Assembly to Kill Mechanism)	1 OF 1	9298418
			NEXT HIGHER ASSEMBLY: 9298417

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Pad missing	Level II	3.2	Visual
102	Hole in pad missing	Level II	3.2	Visual
Minor				
201	Poor workmanship	Level III	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Overlay, Inner	1 OF 1	9298414
			NEXT HIGHER ASSEMBLY: 9298412-1

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material thickness	Level II	3.2	Gage
102	Profile of surface between flange and hole	Level II	3.2	Gage
103	Presence of cracks	Level II	3.2	Visual
Minor				
201	Poor workmanship	Level III	3.15	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Overlay, Outer	1 OF 1	9298413
			NEXT HIGHER ASSEMBLY: 9298412-1

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Profile of surface between flange and hole	Level II	3.2	Gage
102	Depth to bottom of groove (2 places)	Level II	3.2	Gage
103	Material thickness	Level II	3.2	Gage
104	Presence of cracks	Level II	3.2	Visual
Minor				
201	Poor workmanship	Level III	3.15	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH 4.4.2.15	TITLE Spacer, Overlay	SHEET 1 OF 1	DRAWING NUMBER 9298415
			NEXT HIGHER ASSEMBLY 9298412-1

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Diameter of hole	Level II	3.2	Gage
102	Largest outside diameter	Level II	3.2	Gage
103	True position of largest outside diameter with hole	Level II	3.2	Gage
104	Thickness over circular and rectangular protrusions (12 places)	Level II	3.2	Gage
105	Thickness adjacent to protrusions	Level II	3.2	Gage
Minor				
201	Presence of flash	Level IV	3.11	Visual
202	Poor workmanship	Level IV	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Overlay, Welded (Prior to Welding)	1 OF 1	9298421
			NEXT HIGHER ASSEMBLY: 9298412-1

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor 201	Poor workmanship	Level IV	3.15	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.17	Overlay Welded	1 OF 1	9298421
			NEXT HIGHER ASSEMBLY 9298412

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Maximum outside diameter	Level II	3.2	Gage
102	Minimum inside diameter of equator	Level II	3.2	Gage
Minor				
201	Minimum depth of inside diameter	Level II	3.2	Visual
202	Poor workmanship	Level III	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Overlay Assembly (Prior to Trim of Polar Hole)	1 OF 1	9298412-1
			NEXT HIGHER ASSEMBLY: 9298405

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Liquid propellant on exterior of overlay indicates spillage during fill	Level II	3.2	Visual
102	Quantity of liquid propellant incorrect	Level II	3.2	Balance Notes 1 & 3
103	Liquid propellant formulation incorrect			Notes 2 & 3
NOTE:	1. The quantity of liquid propellant shall be determined by weight difference of the overlay before and after loading of the propellant. 2. The liquid propellant shall be lotted and a Propellant Description Sheet (ARRCOM-214-R) shall be prepared for each lot. The propellant shall comply with MIL-P-48333. 3. Alternate: Machine control. Liquid propellant dispensing machines will be qualified at beginning of each production interval, each production hour, and each time the machine is removed from production for maintenance or adjustment. A minimum of two (2) samples will be dispensed and weighted for qualification. Each hour's production or fraction thereof will be accepted or rejected based upon the qualification samples.			

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.19	Overlay Assembly	1 OF 1	9298412-1
			NEXT HIGHER ASSEMBLY 9298405

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Leakage test	Level II	3.2	4.5.9
102	Height of overlay	Level II	3.2	Gage
103	Diameter of hole	level II	3.2	Gage
104	True position of hole with inside diameter at equator	Level II	3.2	Gage
105	Assembly damaged	Level II	3.2	Visual
Minor				
201	Poor workmanship	Level III	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.20	Detonating Cord Assembly (Prior to Assembly to Kill Mechanism & Overlay Assembly)	1 OF 1	9298409
			NEXT HIGHER ASSEMBLY: 9298405

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Cord sheath cracked or broken	Level II	3.2	Visual
102	Booster assembly damaged	Level II	3.2	Visual
103	Kink or sharp bend in cord	Level II	3.2	Visual
Minor				
201	Poor workmanship	Level III	3.15	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Kill Mechanism & Overlay Assembly (Prior to Assembly into Housing)	1 OF 1	9298405
			NEXT HIGHER ASSEMBLY 9366700

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Evidence of leakage of overlay assy	Level II	3.2	4.5.10
102	Height of detonating cord above propellant nut	Level II	3.2	Gage
103	Outside diameter, max. (at equator)	Level II	3.2	Gage
104	Distance across poles of overlay assemblies, max.	Level II	3.2	Gage
105	Grooves in overlays not aligned with groove in propellant nut	Level II	3.2	Visual
106	Detonating cord twisted, kinked, cracked or broken	Level II	3.2	Visual
107	Booster improperly positioned for assembly to S&A	Level II	3.2	Visual
108	Detonating cord not seated in grooves of overlays and propellant nut	Level II	3.2	Visual
109	Sealant inadequate or missing from detonating cord assembly and propellant nut interface	Level II	3.2	Visual
110	Upper seal missing	Level II	3.2	Visual
111	Static functioning test	Level II	3.4	4.5.14 and Table II
Minor				
201	Tape around equator of overlay assemblies missing, loose or not positioned correctly	Level III	3.2	Visual
202	Poor workmanship	Level III	3.15	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.22	Molded Assembly, PDM	1 OF 1	9366661
			NEXT HIGHER ASSEMBLY: 9366700

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Battery Ball height	Level II	3.2	Visual
2	Shorting Bar broken or damaged	Level II	3.2	Visual
3	Temporary Safety Pin missing	Level II	3.2	Visual
4	Temporary Safety Pin over max. (improperly installed)	Level II	3.2	Visual
Major				
101	Battery ball depressed below normal position (below min.)	Level II	3.2	Gage
102	Bare metal exposed in kill mechanism cavity	Level II	3.2	Visual
103	Foreign material in kill mechanism cavity	Level II	3.2	Visual
104	Largest diameter in kill mechanism cavity not prepared for bonding	Level II	3.2	Visual
105	Any external component, including sensor locating plugs, missing or damaged	Level II	3.2	Visual
Minor				
201	Poor workmanship	Level III	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.23	Keeper	1 OF 1	9298404
			NEXT HIGHER ASSEMBLY: 9366700

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Height (.22 - .01) minimum	Level II	3.2	Gage
102	Width (.190 - .005) maximum	Level II	3.2	Gage
Minor				
201	Presence of flash	Level III	3.11	Visual
202	Poor workmanship	Level III	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Mine (Prior to cast filling)	1 OF 1	9366700
			NEXT HIGHER ASSEMBLY: 9366700

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Distance from apex to top of kill mechanism and overlay assembly max.	Level II	3.2	Gage
102	Booster of detonator cord assembly protruding beyond keeper in S&A	Level II	3.2	Visual
103	Keeper not seated on booster	Level II	3.2	Visual
104	Sealing compound improper	Level II	3.2	4.5.11
105	Retaining clips missing or incorrectly positioned	Level II	3.2	Visual
106	Sealing compound not at proper level	Level II	3.2	Visual
Minor				
201	Poor workmanship	Level III	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25	Mine (Prior to assembly of arming hardware)	1 OF 1	9366700
			NEXT HIGHER ASSEMBLY 9366700

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Centerline distance from battery ball to flat per dwg. 9366661	Level II	3.2	Visual
102	Temperature control during cure	Level II	3.2	4.5.13
103	Resin/hardener weight ratio of molding compound	Level II	3.2	4.5.12
104	Compressive strength of molding compound	Level II	3.2	4.5.12
105	Molding compound shelf life expired	Level II	3.2	Visual
Minor				
201	Voids in cavity filled area	Level III	3.2	Visual
202	Poor workmanship	Level III	3.15	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.26	Mine Assembly, M86	1 OF 2	9366700
			NEXT HIGHER ASSEMBLY: ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Safety pin not installed properly	Level II	3.2	Visual
2	Arming lever and cam not in place	Level II	3.2	Visual
3	Safety clip not in place	Level II	3.2	Visual
4	Premature functioning	Level II	3.2	Gage
Special				
A	Mounting plate insulator incorrectly located (see dwg. Not 14 or missing)	SI	3.2	Visual
B	Safety pin washer missing or retaining ring missing	SI	3.2	Visual
C	Shorting bar broken or damaged	SI	3.2	Visual
D	Mounting screws properly installed in sequence	100%	3.2	Visual
E	Hardware mounting surface free of molding compound	100%	3.2	Visual
Major				
101	Distance between battery ball and cam (over max.)	Level II	3.2	Gage
102	Evidence of liquid propellant leakage	Level II	3.2	Visual
103	Any sensor deployed	Level II	3.2	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.26	Mine Assembly, M86	2 OF 2	9366700
			NEXT HIGHER ASSEMBLY: ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major	Cont'd			
104	Any sensor restricted (cap not removed or molding compound flash over sensor cutout)	Level II	3.2	Visual
105	Mounting screws not to proper torque (flush)	Level II	3.2	Gage
106	Shorting bar hook attached to shorting bar and cam per plant	Level II	3.2	Visual
107	Bandoleer defective	level II	3.2	Visual
108	Failed Lot Acceptance Test Function Test (Self Destruct)	36	3.12	4.4.5
Minor				
201	Test pads exposed	Level III	3.2	Visual
202	Poor workmanship	Level III	3.15	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.27	Sleeve, Outer	1 OF 1	9298356
			NEXT HIGHER ASSEMBLY: ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Sleeve, length	Level II	3.2	Gage
102	Sleeve, inner diameter	Level II	3.2	Gage
103	Sleeve, outer diameter	Level II	3.2	Gage
104	Sleeve concentricity	Level II	3.2	Gage
105	Sleeve, chamfer	Level II	3.2	Gage
106	Sleeve, hardness	Level II	3.2	Certification
107	Sleeve, finish	Level II	3.2	Visual
Minor				
201	Mounting plate insulator improperly installed	Level III	3.2	Visual
202	Poor workmanship	Level III	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.28	Explosive Composition A5, prior to loading	1 OF 1	9298358
			NEXT HIGHER ASSEMBLY: 9298358

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Bulk density	Level II	3.7	4.5.2.1
102	Moisture content	Level II	3.7	4.5.1
Minor				
201	Poor workmanship	Level III	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.29	Bag Assembly	1 OF 1	12910117
			NEXT HIGHER ASSEMBLY: 12910121

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
1	Any sensors deployed	SI	3.2	Visual
2	Shorting bar is broken	SI	3.2	Visual
3	Safety pin missing in S&A	SI	3.2	Visual
Major				
101	Bag not clear	Level II	3.2	Visual
102	Presence of heat seal (3 places)	Level II	3.2	Visual
103	Notches missing or improperly formed	Level II	3.2	Visual
104	Mines not identified		3.2	Visual
105	Notches penetrating heat seal area, 2 places	Level II	3.2	Visual
Minor				
201	Poor workmanship	Level III	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.30	M86 Apers Mine with Bandoleer	1 OF 1	12910121
			NEXT HIGHER ASSEMBLY: ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Bandoleer assembly not marked correctly	Level II	3.2	Visual
Minor 201	Poor workmanship	Level III	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.31	Packing and Marking for Shipping & Storage Container, PA19 for Mine, Apers M86, Classroom Model, Special Test Mine Demonstration Mine	1 OF 1	9366709
			NEXT HIGHER ASSEMBLY 9366710

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
1	Marking on PA19 container incorrect	SI	3.2	Visual
Major				
101	Packing material(s) incorrect sizes	Level II	3.2	Visual
102	Moisture in packing material above max.	Level II	3.2	Visual
103	Outer barrier bag marked incorrectly	Level II	3.2	Visual
104	Mine inserted incorrectly into inner bag and bandoleer	Level II	3.2	Visual
105	Outer bag loaded incorrectly	Level II	3.2	Visual
106	Outer bag sealed incorrectly	Level II	3.2	Visual
107	Pull tabs missing from outerbag	Level II	3.2	Visual
108	PA19 container not loaded correctly	Level II	3.2	Visual
Minor				
201	Poor workmanship	Level III	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.32	Box, Wirebound	1 OF 1	9366708
			NEXT HIGHER ASSEMBLY 9366710

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Incorrect material	Level II	3.2	Visual
102	Inside tolerances incorrect	Level II	3.2	Visual
Minor				
201	Poor workmanship	Leve IV	3.15	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-70696A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.33	Packing and Marking for Box, Wirebound for Mine, Apers M86, Classroom Model, Special Test Mine or Demonstration Mine	1 OF 1	9366710
			NEXT HIGHER ASSEMBLY 9366711

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Packing material missing	Level II	3.2	Visual
102	Incorrect packing material used			
103	Moisture content of packing materials over max.	Level II	3.2	Visual
104	Packing material installed incorrectly	Level II	3.2	Visual
105	Marking incorrect or missing	Level II	3.2	Visual
106	Wirebound box closed incorrectly	Level II	3.2	Visual
Minor				
201	Poor workmanship	Level III	3.15	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph/Inspection Method" column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval, inspections equipment designs in accordance with the terms of the contract. See Section 6 or MIL-A-48078.

4.4.5 Lot acceptance testing (LAT). The contractor shall perform LAT testing on HE mines for each lot shipped. The procedure shall be as follows:

a. The HE mine shall be placed into a remote arming and deploying fixture.

b. Pulling off of the safety clip will be recorded while using a pull gage.

c. Once the mine starts to arm (battery ball depressed) the mine should be deployed immediately. Timing of the mine should commence to insure that the detonator, piston actuator or the gas generator didn't function before 25 seconds after initiation of the arming cycle.

d. The mines shall be deployed in such a manner that self destruct monitoring can be accomplished.

e. If a mine functions prior to 80% of self destruct time, it shall be classified as an early out (Eo) defect. If the mine functions after 100% of the self destruct time it shall be called a hazardous dud (HD) defect. The lot shall be rejected if 4% of the test samples fail for self destruct or low order (Lo) and 35 for early out (Eo).

f. The test samples shall be temperature conditioned for a minimum of 4 hours as follows:

(1) 12 samples at a temperature of $-25^{o}F + 5^{o}F$
(2) 12 samples at a temperature of $+70^{o}F + 5^{o}F$
(3) 12 samples at a temperature of $+120^{o}F + 5^{o}F$

After conditioning, the samples may be tested at the ambient temperature within 5 minutes from the time they are removed from the temperature conditioned box.

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4.4.5.1 Hazardous dud (HD) evaluation. Any mine which has not functioned in any of the function mode groups prescribed shall be subjected to the hazardous dud evaluation procedure within five hours after the expiration of the design time out period. This procedure shall be as follows:

a. One deployed tripline shall be pulled. if there is not function, the AD procedures shall be attempted. If the mine still has not functioned, the mine shall be left to SD.

b. After completion of the prescribed hazardous dud procedures steps, all remaining triplines shall be cut three (3) inches from the sensor cavity.

After completion of the prescribed hazardous and procedure steps, mines shall be placed in a secure area and be allowed to time out to 125% of maximum SD time. Time starts when the mines are physically placed in the secure area. Following time-out, mines shall be examined for evidence of detonation function (STM only 9366705). Any mines which have functioned during the time-out period shall be classified as hazardous duds and so recorded.

4.4.5.2 Scoring.

a. All test data shall be recorded.

b. Prior to testing a temperature/function sub lot the field will be cleaned of all residual test hardware.

4.4.5.3 Data recording. Data sheets used to record test data and information provided in final Test Record, will be structured to contain as a minimum.

a. Mine Lot Numbers.

b. Climatic conditions of the day of test.

c. Mine sample number and associated function mode.

d. Initial mine orientation (face identification and detailed identification of faces) encountered during movements in anti-disturbance testing.

e. Firing train doesn't propagate the HE (low order detonation).

f. Mine detonates before 25 seconds (premature detonation).

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g. Number of earlies (EO) and elapsed time early occurred (if known) by mine number.

h. Deployment of sensors prior to 25 seconds (premature arming) or after 75 seconds.

i. Defective arming mechanism (dud).

j. Mine functioning times (self destruct).

k. Hazardus Dud Mine number and time of functioning (as a %) after battery activation.

4.4.5.4 Packaging or marking problems. First Article of Lot Acceptance/rejection Criteria - the lot shall be rejected if the results reveal any of the following defects:

a. One or more hazardous duds.

b. Safe Dud Self-Destruct rate plus low orders exceeds three percent (3%) of self destruct sample.

c. Early Out Self-Destruct rate exceeds three percent (3%) of self destruct sample.

4.5 Methods of inspection.

4.5.1 Moisture content of explosive composition A5 (for loading upper and loading shell assemblies). The contractor shall provide adequate controls to insure compliance with the applicable requirement. The moisture content of the explosive composition shall be determined at the loading station at the time of loading. This determination shall be made at least once each either (8) hours of production. Sampling of the explosive composition shall be in accordance with MIL-E-14970. In addition, if the moisture content fails to comply with the requirement, and loading has not begun, the explosive composition shall be rejected until corrective action has been taken as verified by repeating the moisture content determination. If loading has begun, all assemblies loaded with the non-conforming explosive composition shall be rejected. The moisture content shall be determined in accordance with MIL-E-14970.

4.5.2 Density of explosive composition A5.

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4.5.2.1 Bulk density of explosive composition. The contractor shall provide adequate controls to insure compliance with the applicable requirements. The bulk density of each batch of the explosive composition shall be determined at the loading station at the time of loading. This determination shall be made on each bucket on each loading machine immediately prior to its use. Sampling shall be in accordance with MIL-E-14970. In addition, if the bulk density fails to comply with the requirement specified on the applicable drawing and loading has not begun, the composition lot shall be rejected until corrective action has been taken as verified by repeating the bulk density determination. If loading has begun, assemlies with non-conforming composition shall be rejected. The following procedure shall be used to determine the bulk density of the explosive composition.

BECHTEL METHOD OF BULK DENSITY MEASUREMENT

a. Use 100cc capacity cup (metal cup recommended), polished inside and outside to eight (8) micro-inch finish and a tare weight equal to the weight of the empty cup. The dimensions of the 100cc cup shall be:

I.D. = 4.0 cm Inside depth = 7.958 cm.
Inside bottom radius = 0.076 mm, max.
Wall thickness = 0.2 cm (approx.)

b. Place both 100cc cup and the tare weight on analytic balance scale, or equal, one on each pan and check that cup and tare weight balance.

c. Set cup on level surface and fill with composition to be bulk-density tested. To fill, use a scoop, approximately tablespoon size, or a tablespoon, and fill the cup above the rim. Do not tamp or shake at any time. Level off excess composition with a straight edge, keeping side of straight edge vertical to rim of cup.

d. Place filled cup on balance pan. Add gram weights to other pan with tare weight until balance is reached. Number of grams added will give the bulk density in percent.

4.5.2.2 Consolidation density of charge in upper and lower shell assemblies. A minimum of two (2) samples each, representative of both the shell tube assembly and the lower shell being processed, shall be loaded and the densities determined to qualify each press prior to its use in the loading process in production. In addition, a sample of two (2) assemblies shall be randomly selected from the last portion of each one (1) hour's production from each press and the density determined as a control of each press during production.

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If any defective assemblis are found when inspection is performed for control of a press, the press producing the defective assembly shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All presses removed from production because of some fault, may, after corrective action, be returned to production, providing they pass the qualification test above. The following procedure shall be used to determine the density of the explosive composition.

When weighing in water, use distilled water. A suitable wetting agent may be used.

a. Prior to loading, weigh the lower shell or upper shell as applicable.

b. Prior to assembly, weigh the same component in water (grams).

c. Thoroughly dry the component from b. (above) and assemble with explosive composition using normal production methods.

d. Weigh the loaded component in air (grams).

e. Weigh the same loaded component in water (grams).

f. Calculate the weight of the explosive composition in air (d) - (a) (grams).

g. Calculate the weight of the explosive composition in water (a) - (b) (grams).

h. Calculate the weight (and volume) of water displaced by the explosive composition (f) - (g) (grams x lcc per gram).

i. Calculate the density of the loaded explosive composition (f)/(h) (grams per cc).

4.5.3 Moisture content of propellant blend for propellant nut assembly. The contractor shall provide adequate controls to insure compliance with the applicable requirements. The moisture content of the propellant shall be determined at the loading station the time of loading the propellant nut assemblies. This determination shall be made at least once for each twenty-four (24) hours of continuous production and at least once for each lot of propellant nut assemblies. Sampling and testing of the propellant shall be in accordance with Method 101.1.2 of MIL-STD-286. If the moisture

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content fails to comply with the requirement specified on the applicable drawing and loading has not begun, the propellant shall not be used until corrective action has been taken as verified by repeating the moisture content determination. If loading has begun, assemblies loaded with the non-conforming propellant shall be rejected. Method T.03.5 of MIL-STD-286 may also be used for this test.

4.5.4 Security of orifice plate in propellant nut assembly. At the start of production each day, each machine to be used for staking the orifice plate in the propellant nut shall be used to assemble two (2) inert (empty) propellant nut assemblies to qualify each machine for production. In addition, two (2) inert (empty) assemblies per machine shall be assembled after each hour's production or portion thereof, as a control of the qualification of the equipment, the equipment producing the defective assembly shall not be used in production. If any defective assembly is found when inspection is performed for control of the equipment, the equipment which produced the defective assembly shall be removed from production. In addition, that portion of propellant nut assemlies produced by the equipment since the last check shall be rejected and returned to the contractor for corrective action.

All equipment removed from production because of some fault, may, after corrective action, be returned to production providing it passes the qualification test above.

The inert (empty) assemblies shall be tested by drilling a 0.110 inch diameter hole (#35 drill) through only the head of the propellant nut and through a .109 - .005 inch diameter rod to the adhesive side of the orifice stakes after the specified force has been applied, the assembly shall be classed defective.

Alternate method: A portion of the head of the propellant nut shall be removed by milling or drilling and the propellant shall be removed. The specified force shall be applied through a 0.109 - 0.005 inch diameter rod to the adhesive side of the propellant nut at any of the stakes after the specified force has been applied, the assembly shall be classed defective.

4.5.5 Qualification test of delay detonator serial lot and propellant nut assembly serial lot.

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4.5.5.1 Test procedure. Prior to assembly of the Kill Mechanism and Overlay Assembly lot, delay detonators and propellant nut assemblies from the respective serial lots to be used in the kill mechanism and overlay assembly lot shall be subjected to this test. The test shall be performed using the method specified on Dwg. 9298699. Twenty (20) of the samples shall be tested at -25 ± 5°F, twenty (20 of the samples shall be tested at 125 ± 5°F and ten (10) of the samples shall be tested at 70° ± 5°F. The propellant nut assemblies and delay detonators shall be conditioned at the specified test temperatures for 2 hours minimum prior to testing. All test data shall be recorded. This test is for information only. The results of the contractor tests shall be held for Government review.

4.5.2 Adequacy of sealant of kill mechanism assembly. At the start of each shift's operation, five (5) completed kill mechanism assemblies shall be selected from each piece of equipment for determination of sufficiency of sealant indicated by the read not less than 1/32 of an inch wide at contact of ball halves, 360° around, or as indicated upon disassembly by sealant completely covering the surface of the component in an area between the outer edge and three sixteenths (3/16) to five sixteenths (5/16) of an inch toward the center for qualifying the equipment. In addition, two (2) assemblies per machine shall be selected from each hour's production, or portion thereof, for determination of sufficient sealant as a control of the equipment. If any defective assemblies are found during qualification of the equipment, the equipment producing the defective assemblies shall not be used in production. If any defective assembly is found when inspection is performed for control of the equipment, the equipment which produced the defective assembly shall be removed from production. Further, that portion of production since the last equipment check, shall be returned to the contractor for corrective action.

All equipment removed from production because of some fault, may, after corrective action be returned to production, providing it passes the qualification test above.

4.5.3 Minimum assembly torque of propellant nut assembly. In place of the normal sampling procedures associated with the classification of defects and with the approval of the Contracting Officer, the following inspection procedure may be used: at the start of each day, and whenever a new piece of equipment is put into operation, five (5) kill mechanism assemblies shall be selected from each piece of equipment for determination of minimum assembly torque of the propellant nut assemlies for qualifying the equipment. in addition, five (5) kill mechanism assemblies per machine shall be

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selected from each hour's production or fraction thereof, for the determination of the minimum propellant nut assembly torque as a control of the equipment. If any defective mechanism assembly is found when produced the defective assembly shall be removed from production. Further, that portion of production since the last equipment check shall be returned to the contractor for corrective action. All equipment removed from production because of some fault, may, after corrective action, be returned to production providing it passes the qualification test above.

4.5.4 Control of quantity of propellant in overlay assembly. The contractor shall provide necessary controls to assure that the quantity of propellant loaded into each overlay assembly is within the requirement specified on the applicable drawing. The method of control shall be submitted for approval as specified in 4.4.4.

4.5.5 Leakage test of overlay assembly. The assemblies shall then be cleaned in an acetone bath and placed "equator down" on filter paper and filter paper placed on top of overlay assembly "polar hole" in a tray. The assemblies shall be subject to a vacuum of 10mm mercury (0.193 psia) for five minutes minimum and visual observation made for propellant leakage as evidenced by orange staining on the filter paper. The assemblies shall then be cleaned in an acetone bath and place "equator down" on filter paper and filter paper placed on top of the overlay assembly "polar hole" in a tray. The assemblies shall then be placed in an oven at 160 $\pm$ 5°F for forty (40) hours minimum, and visually examined for any evidence of propellant leakage as evidenced by orange staining on the filter paper. All assemblies which have any evidence of leakage shall be classed defective and removed from the lot.

4.5.6 Leakage test of kill mechanism and overlay assembly. The assemblies shall be tested by conditioning in a preheated oven at 160 $\pm$ 5°F for 15 minutes minimum. The assemblies shall then be removed from the oven and immediately subjected to a vacuum of 10MM mercury absolute pressure (0.193 psia) for 5 minutes minimum. After removal from the vacuum chamber the assemblies shall be visually inspected for evidence of liquid propellant leakage especially around the pole and equator welds of the overlay assemblies. Assemblies which show any evidence of leakage including staining of the tape around the equator, shall be classed defective and removed from the lot.

4.5.7 Quality controls for sealing compound.

4.5.7.1 Sealing compound batch. A batch of sealing compound is herein defined as that quantity of material mixed at one time for use in assembling kill mechanism and overlay assemblies in housings prior to cast molding.

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4.5.7.2 Certification of properties. Certification of compliance with the requirement for gel time, shall be available for each batch of sealing compound. In addition, the data from the test shall be furnished to the Government upon request. If Part A is mixed and stored, the tests shall be run after storage and immediately prior to use.

4.5.7.3 Records of addition of catalyst (part B). After the compound has been mixed and the samples for gel time have been taken records shall be maintained to show the amount of catalyst added and any other additions or adjustments made to the compound at any time until the batch is consumed.

4.5.7.4 Records of shelf life. When the compound is mixed and stored prior to use, containers shall be marked to show the date the compound was mixed and a date before which the compound must be used. In addition all containers with the ingredients for the sealing compound shall be marked with the appropriate shelf life dates. Pre-mixed "Kits" of ingredients shall also be marked with appropriate shelf life dates. Shelf life expiration dates of pre-mixed kits shall be based on the ingredient(s) with the shortest shelf life or the earliest expiration date whichever is earlier.

4.5.8 Quality controls for each molding compound batch for casting housings/assemblies or assembling plugs to housing.

4.5.8.1 Molding compound batch definition. A batch of molding compound is herein defined as that quantity of compound mixed at one time using one combination of base resin and hardener lots from one manufacturer that have been tested together and met the requirements of MIL-M-48245 and which is used to assemble housings/assemblies in one unchanged process. When a continuous type mixing is used, a batch is defined as that quantity of molding compound mixed and dispensed in one continuous unchanged process during a four (4) hour period or portion thereof using one combination of base resin and hardener lots from one manufacturer that have been tested together and met the requirements of MIL-M-48245.

4.5.8.2 Resin/hardener weight ratio control.

a. Hand mixing: For each batch of molding compound mixed at one time, the hardener and resin shall be weighed separately to verify that the ratio is between 1.8/1 and 2.2/1 before use.

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b. Automatic mixer method: At the beginning of production and once each four (4) hours during production two test shots shall be drawn and with the mixer set to expel the hardener and resin separately. The hardener and resin shall be weighed to verify that the ratio is between 1.8/1 and 2.2/1. If the shots fail to meet the requirement, all wedges filled since the last acceptable test shall be rejected. Further, the mixer shall be corrected and shall pass the above test before being returned to use.

4.5.8.3 Quality controls for each molding compound batch for casting housings/assemblies. Two test bars from each batch of molding compound shall be molded at the same time the batch is used to assemble housings. The test bars shall be 0.5 inch diameter by 1.0 inch long or 0.5 inch square by 1.0 inch long. Test bar molding shall be by best commercial practice, including vibration and evacuation if necessary. The mix weight ratio limits shall be 2/1 ± 5% (1.8/1 to 2.2/1). The test bars shall be processed with the items assembled from the batch of molding compound which the bars represent., i.e. the bars shall be cured in the same ovens and subjected to the same temperatures for the same time periods as the items they represent. After completion of the cure, one bar shall be tested for ultimate compressive strength as specified in ASTM D-695. The ends shall be flat and parallel to within .003 inch, and shall be perpendicular to the long axis of the test bar. Speed of testing shall be as specified in the ASTM. If the results of this test are below the limit specified in 3.9, the munitions represented by the bar shall be rejected. However, if the test bar failure can be attributed to a void, which is verified by the Government inspector, the second test bar (cast at the same time and representing the same batch of assemlies) shall be tested as a replacement.

4.5.9 *Temperature control during molding/cure operations*. Whenever the housing or munition is subjected to heating in ovens, temperature records shall be maintained and reviewed to assure that the minimum temperature has been reached and that the maximum temperature has not been exceeded (See 6.5). If any units have been subjected to temperatures beyond the maximum limit at any time, they shall be rejected. Temperature measuring instruments shall be approved and calibrated in accordance with 4.4.4.

In addition, if units are stored where temperatures may exceed the limit specified in 3.10, temperature monitoring equipment shall be provided and constant records maintained of the temperature to assure that the temperature limit is not exceeded.

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4.5.10 Static functioning test of kill mechanism and overlay assembly. The sample kill mechanism and overlay assembly (KM and O) shall be assembled into an ADAM firing train housing, or reject ADAM firing train housing, provided KM&O can be properly seated. The KM&O assembly shall be assembled into the housing, timing, fusing (H.T.F.) housing using normal assembly techniques. The booster cup of the detonating cord assembly shall be aligned with, and positioned against the output side of the lead cup assembly in the S&A. The second detonating cord assembly shall be placed in the same position as the first, resting directly on top and the cord shall bend 90° upward, protruding vertically out of the area to be closed and bonded. The booster keeper and the retaining clips shall be assembled. The H.T.F. housing shall then be closed, using normal assembly procedures. After curing, those H.T.F. assemblies to be tested at temperature extremes shall be temperature conditioned for two hours minimum at the temperature specified in Table II. The H.T.F. assemblies shall be tested in the orientations and the temperatures specified in Table II. The testing procedures in the following paragraphs shall apply unless otherwise specified by the Contracting Officer. Test procedures and equipment designs shall be documented and submitted for approval (see 4.4.4 and 6.3).

4.5.10.1 Test arena. The arena or room shall be not less than twenty feet square and fifteen feet in height. It shall be constructed of material that will resist the blast of twenty-five areas where fragments, parts, etc., could lodge and be lost. At least one wall shall be positioned with a viewing port set at a height above the test surface to photograph item functioning in front of a suitable grid for subsequent examination and evaluation. Alternate construction may be used with the approval of the contracting officer.

Provisions shall be made to determine the velocity of the kill mechanism by determining the distance it travels and measuring the time from mine breakup to the audible report from kill mechanism functioning.

4.5.10.2 Firing lines. A set of Firing lines shall be provided and they shall be expendable if damaged. Undamaged lines may be reused for subsequent tests.

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4.5.10.3 Firing equipment. The firing equipment shall:

a. Be capable of checking the continuity of the firing lines.

b. Provide for control of all camera stations and for determining elapsed time.

4.5.10.4 Test procedures. The H.T.F. assemblies shall be placed on a test pad of dry sand (sand capable of free flow at time of test). The firing line set shall be shorted. An M100 detonator shall be placed on the protruding detonating cord assembly, the firing line set will be attached to the detonator. The operator shall then exit from the test arena.

Following completion of all procedures for preparation of the H.T.F. assembly functioning shall be accomplished using the following procedures.

The camera shall be positioned so that its field of view will record the entire functioning sequence of the H.T.F. assembly. The camera and firing shall then be connected to the firing equipment. The electric detonator shall be fired. Any H.T.F. assembly which fails to function the KM&O assembly due to the detonator failure, breakage of detonating cord assembly shall be considered a no test item and a replacement H.T.F. assembly shall be tested in its place.

4.5.10.5 Test evaluation (see 3.4). The photograph shall be examined to determine compliance with the requirements of 3.4. All residue from the test shall be collected. If any large fragments of the kill mechanism are found which exceed the dimensional value specified in 3.4 in any direction, the KM & O shall be considered to have failed the test. All test results and any other pertinent information gathered during the test shall be forwarded as specified by Data Item Description DI-R-1724. Such information shall include:

a. Height of functioning of main charge (see 3.4c).

b. Fragmentation (compliance or non-compliance with 3.4d).

c. Average height of burst for the sample (see 3.4d).

d. A record of the "no test" test housings including the failure mode, e.g. piston actuator, detonator, or test equipment.

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e. A record of any unusual visual or audible conditions when the main charge functions. (e.g. yellow or white cloud indicating low order function vs black cloud indicating high order function).

f. Velocity of the kill mechanism assembly.

5. PACKAGING

5.1 Unit packaging.

5.1.1 Level A. Level A unit packaging shall be in accordance with Dwg. 9366709.

5.1.2 Level B. Level B unit packaging shall have the same requirements as for Level A.

5.1.3 Level C. Level C unit packaging shall have the same requirements as for Level A.

5.2 Packing.

5.2.1 Level A. Level A packing shall be in accordance with Dwg. 9366710.

5.2.2 Level B. Level B packing shall have the same requirements as for Level A.

5.2.3 Level C. Level C packing shall have the same requirements as for Level A.

5.3 Marking.

5.3.1 Level A. Level A marking shall be in accordance with Dwg. 9366710.

5.3.2 Level B. Level B marking shall have the same requirements as for Level A.

5.3.3 Level C. Level C marking shall have the same requirements as for Level A.

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5.4 Shipping. When shipment of units from more than one lot are made at one time, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit. The Mine, Anti-Personnel, M86 shall be shipped Category II munitions of Arms, Ammunition, and Explosives (AA&E).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The assemblies covered by this specification are intended for use in a Mine, Anti-Personnel, M86.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced in Section 2.1.

c. Requirements for submission of first article sample.

6.3 Consideration of data requirements. The following data requirements should be condisered when this specification is applied on a contract, per the contract Data Requirements List, DD Form 1423.

6.4 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: AMSMC-QAT-M, Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the Contract.

6.5 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conforamnce provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.6 Documentation of molding process controls. The contractor's documentation of molding process shall specify, as a minimum, controls and procedures for the following:

a. Confirmation of accuracy of temperature-measuring instruments, including the method, location and frequency basis for obtaining temperature checks of all the temperature recording instruments for all the molding and curing phases.

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b. Homogeneity of molded munition, to preclude mixing of munitions from different lots during molding and cure.

c. The storage of permanent records, available for Government review, of the time-temperature cycle for all phases of the molding and cure process and for the items in a and b above. The records shall provide suitable identification by molding compound batch and lot number as applicable.

6.7 Bulk density of explosive composition A5. The purpose of maintaining the bulk density of the explosive composition A5 at 0.83 to 1.00 is to remove any excessively fine or coarse particles when present in certain batches. Excessive concentrations can cause low order function.

The bulk density requirements can be met by screening the powder using a Standard US Sieve #20 over #50 US Sieve. Only the powder particles that are retained on the #50 sieve are used. The powder that falls through the #50 sieve is considered too fine and is scrapped. The powder retained on the #20 sieve is considered too coarse and is also scrapped.

It is recommended that the particle size distribution of each batch of powder be determined before and after the above sieves #20, 30, 40, 50, 60, 70, 80, 100, 140 and 200 and that the data obtained be included with the bulk density certification.

6.8 Safety precautions for molding compound and sealing compound. When handling, mixing or curing the molding or sealing compound or its constituents, suitable precautions must be taken, as some of the constituents and compounds are toxic and may exude toxic fumes when mixed or heated. Use adequate ventilation. Avoid contact with skin, eyes and clothing and prolonged repeated breathing of fumes.

6.9 Toxicity of liquid propellant. See MIL-P-48333.

6.10 Data cards. In addition to the normal distribution of records, one copy of the ammunition data card for each munition lot shall be forwarded to: Commander, USARDEC, ATTN: SMCAR-QAT-M, Picatinny Arsenal, New Jersey 07806-5000.

6.11 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Engineering Center, (ARDEC), may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

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6.12 Distribution of ammunition data cards, inspection and test results, process controls. Distribution of ammunition data cards, inspection and test results, process controls shall be sent to:

Commander
ARDEC
ATTN: SMCAR-QAT-M
Picatinny Arsenal, New Jersey 07806-5000

6.13 Subject term (key word) listing.
Composition A-S
Liquid Propellant
Bandoleer

6.14 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian
Army-AR

Preparing activity
Army-AR

(Project: 1345-A636)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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3. DOCUMENT TITLE: MINE, ASSEMBLY, ANTI-PERSONNEL, M86, LAP

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
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c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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